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CBCS SCHEME

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BEC302

Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Digital System Design Using Verilog

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Develop a truth table of logic which takes two , 2 – bit binary numbers as its input and generate on output equal to 1, when the sum of the two numbers is odd.	7	L1	CO1
	b.	Convert the following Boolean function into : i) $f(abc) = (\bar{a} + b)(b + \bar{c})$ – Min term canonical form. ii) $f(xyz) = x + \bar{x}\bar{z}(y + \bar{z})$ – Max term canonical form.	7	L1	CO1
	c.	List the difference between Prime implicant and Essential prime implicant.	6	L1	CO1
		OR			
Q.2	a.	Simplify the given Boolean function using Quine Mc Cluskey Minimization Technique for the function $R = f(abcd) = \Sigma(0, 1, 2, 6, 7, 9, 10, 12) + dc(3, 5)$.	10	L1	CO1
	b.	Find the minimal sum and minimal product for the given function using K – map method for the function $R = f(abcd) = \Sigma m(0, 1, 3, 7, 8, 12) + dc(5, 10, 13, 14)$.	10	L1	CO1
		Module – 2			
Q.3	a.	List the difference between decoder and encoder and implement full adder using IC – 74138.	10	L3	CO2
	b.	What is Comparator? Design a 2 – bit digital comparator.	10	L3	CO2
		OR			
Q.4	a.	Realize the Boolean function $P = f(wxyz) = \Sigma(0, 1, 5, 6, 7, 10, 15)$ using i) 8 : 1 MUX ii) 4 : 1 MUX.	10	L2	CO2
	b.	With neat logic diagram, explain carry ahead adder.	10	L2	CO2
		Module – 3			
Q.5	a.	Explain the working of master slave JK flip flop with help of Logic diagram , Function table , Logic symbol and Timing diagram.	10	L1	CO3
	b.	Obtain the characteristic equation for : i) SR flip - flop ii) J – K – flip - flop iii) D – flip - flop iv) T – flip - flop.	10	L2	CO3
		OR			

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Q.6	a.	Design a Synchronous 3 – bit up counter using J K – flip - flop.	10	L4	CO3
	b.	Design a 4 – bit universal shift register using positive edge triggered D – flip - flop and 4 : 1 MUX , to operate as shown in table below :	10	L4	CO3

S_1	S_0	Register Operation
0	0	- Hold
0	1	- Shift right
1	0	- Shift left
1	1	- Parallel load operation

		Module – 4			
Q.7	a.	Illustrate the structure and verilog module and write a verilog code for Half – adder using structural model.	10	L3	CO4
	b.	What are different types of operators used in HDL with example?	10	L2	CO4
		OR			
Q.8	a.	Illustrate the structure of Data flow description with example.	10	L3	CO4
	b.	Write the syntax of conditional signal assignment statement. Write a code for 4 : 1 MUX using conditional signal statement.	10	L2	CO4
		Module – 5			
Q.9	a.	Write the structure of Verilog behavioral description.	6	L2	CO4
	b.	Write the syntax of IF statement with example.	7	L2	CO4
	c.	Write a code for D – Latch using Behavioral description.	7	L2	CO4
		OR			
Q.10	a.	Write the syntax of While loop statement with example.	10	L2	CO4
	b.	Write a verilog code of a 3 – bit ripple carry adder using Structural description method.	10	L2	CO4

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